Hostel Accommodation & Student Services

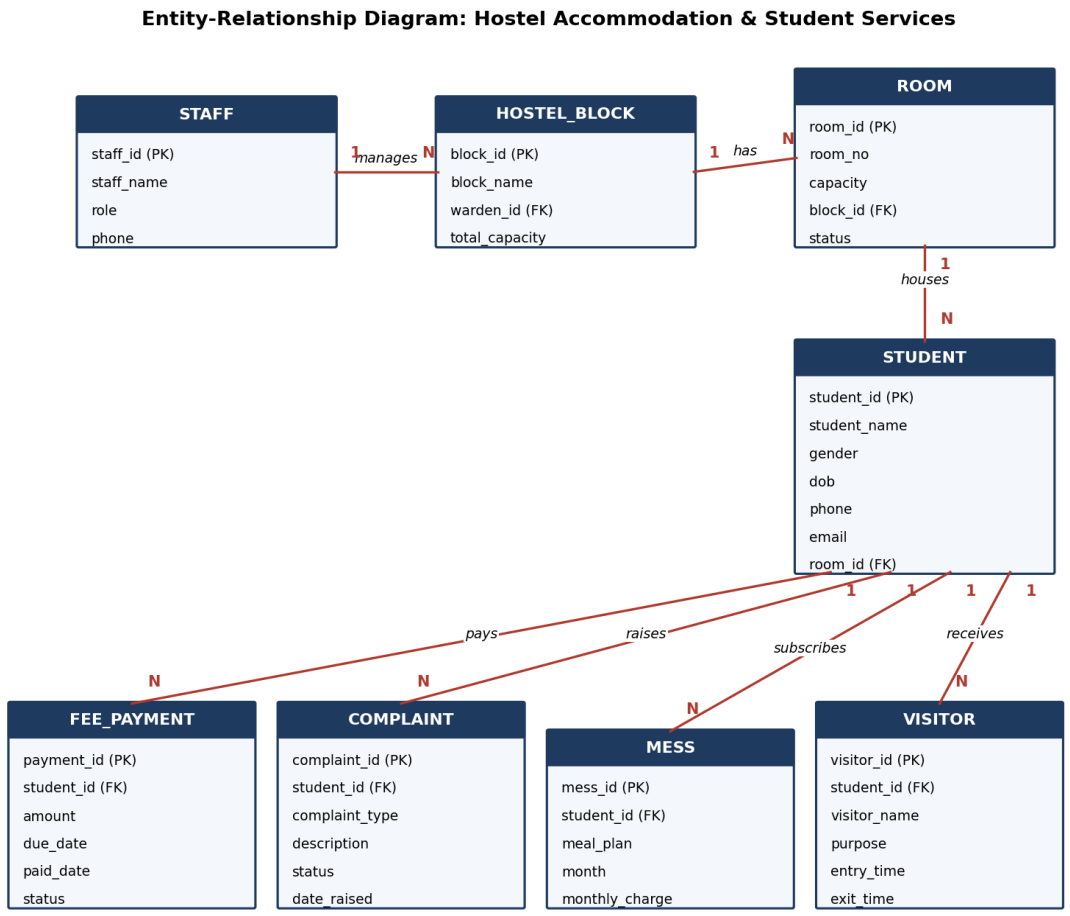
Presentation-III: Final Demonstration

Name: [Hussain Shaik] Roll no.: [25WU0102100]

Problem and Objectives

- Paper registers cause duplicate records, lost fee data and slow complaint handling
- Objective: one normalised MySQL database for staff, blocks, rooms, students, fees, complaints, mess and visitors
- Web admin panel to insert, view and delete records in all 8 modules
- SQL reports: room occupancy, unpaid fees, open complaints, mess revenue

ER Diagram: 8 Related Tables

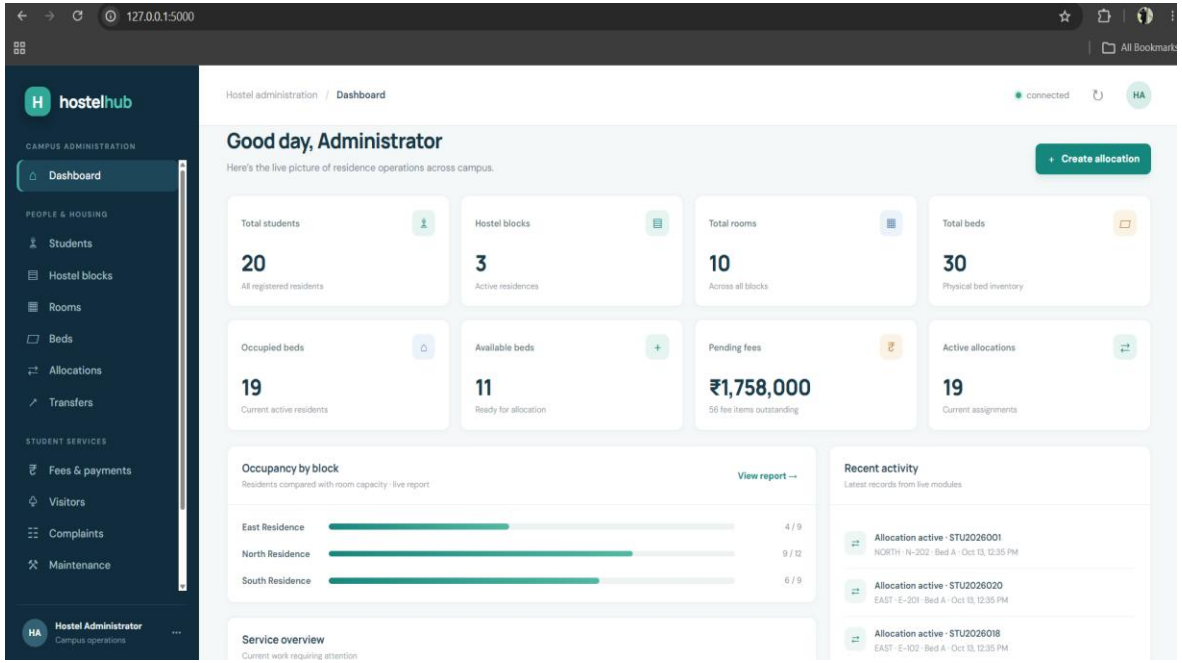


- Staff 1:N Hostel_Block (warden)
- Hostel_Block 1:N Room
- Room 1:N Student
- Student 1:N Fee_Payment, Complaint, Mess, Visitor
- Schema is in 3NF; foreign keys keep data consistent

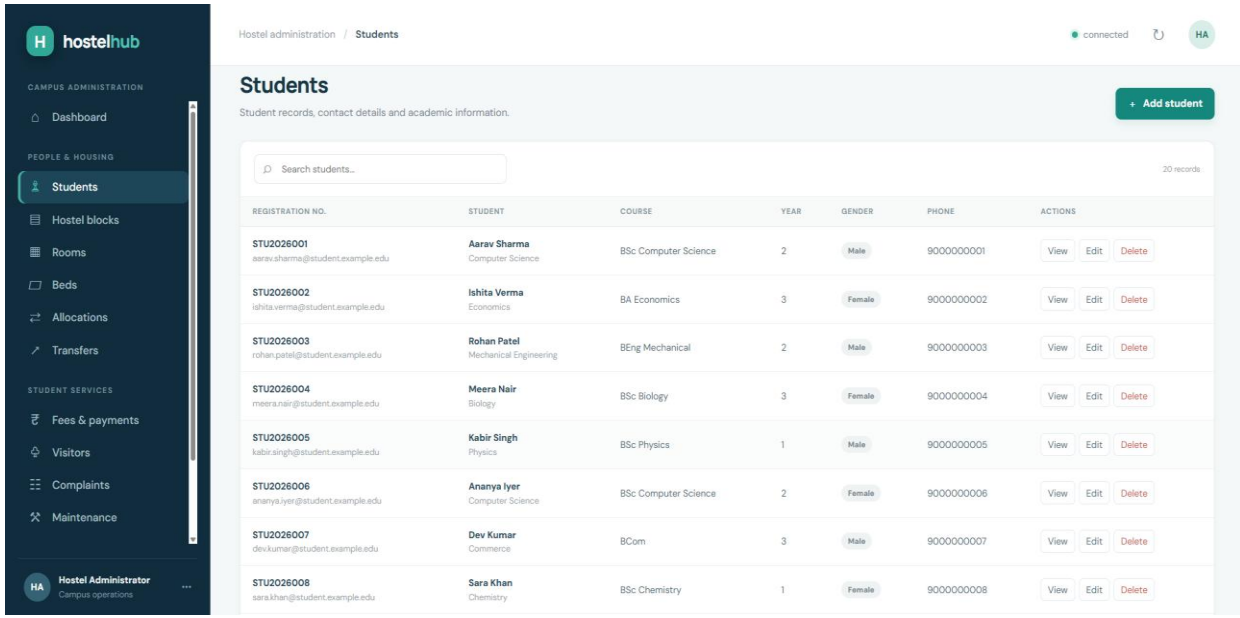
Technology and Implementation

- Node.js v24.15.0, started with npm start (port 3000)
- MySQL database; connection pool in config/db.js
- HTML, CSS, JavaScript panel with sidebar for 8 modules
- Parameterised queries (?) prevent SQL injection
- Drop-downs filled from related tables enforce valid foreign keys

UI Screens: Dashboard & Students Management Screen

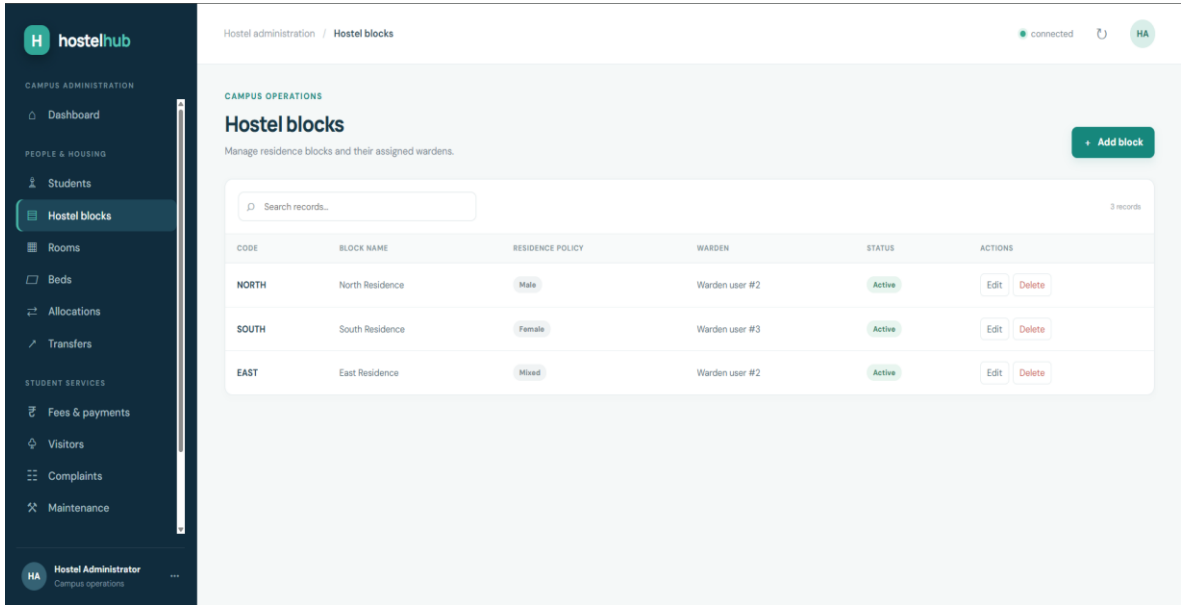


Dashboard

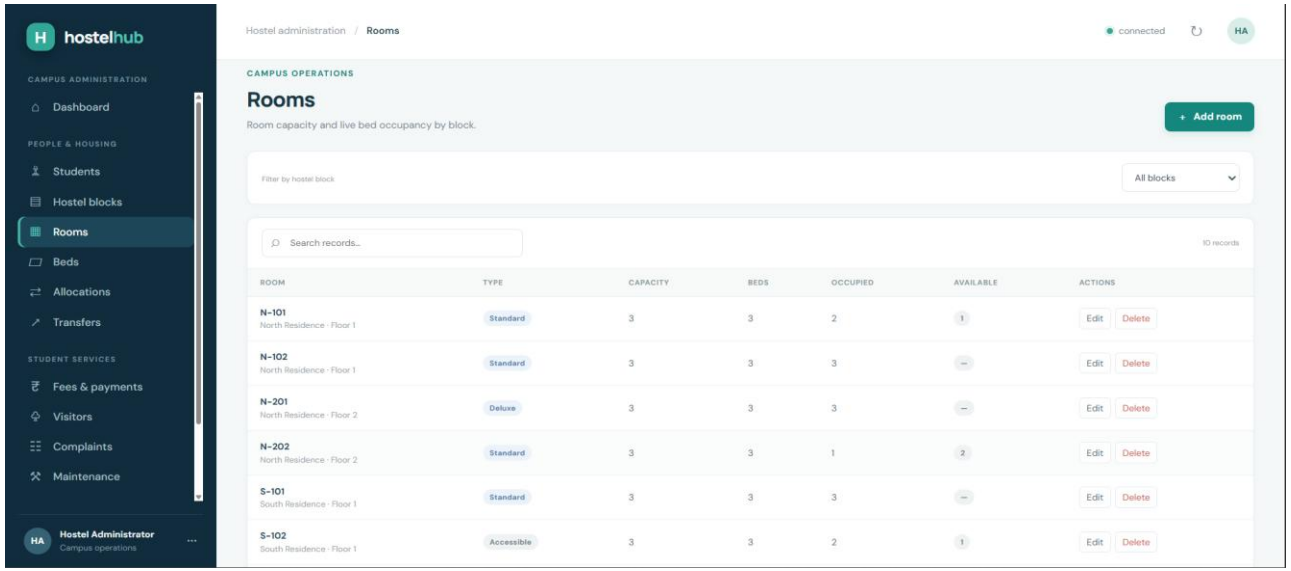


Students Management Screen

UI Screens: HOSTEL BLOCKS AND ROOMS

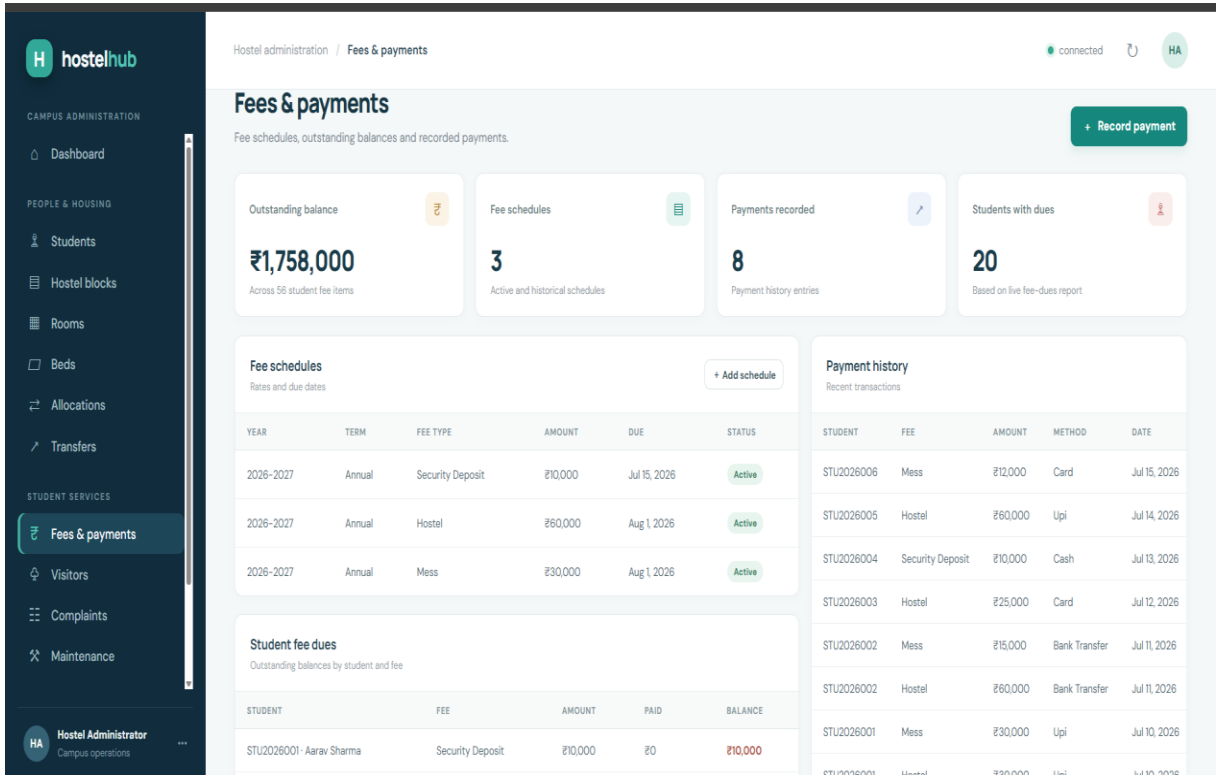


HOSTEL BLOCKS

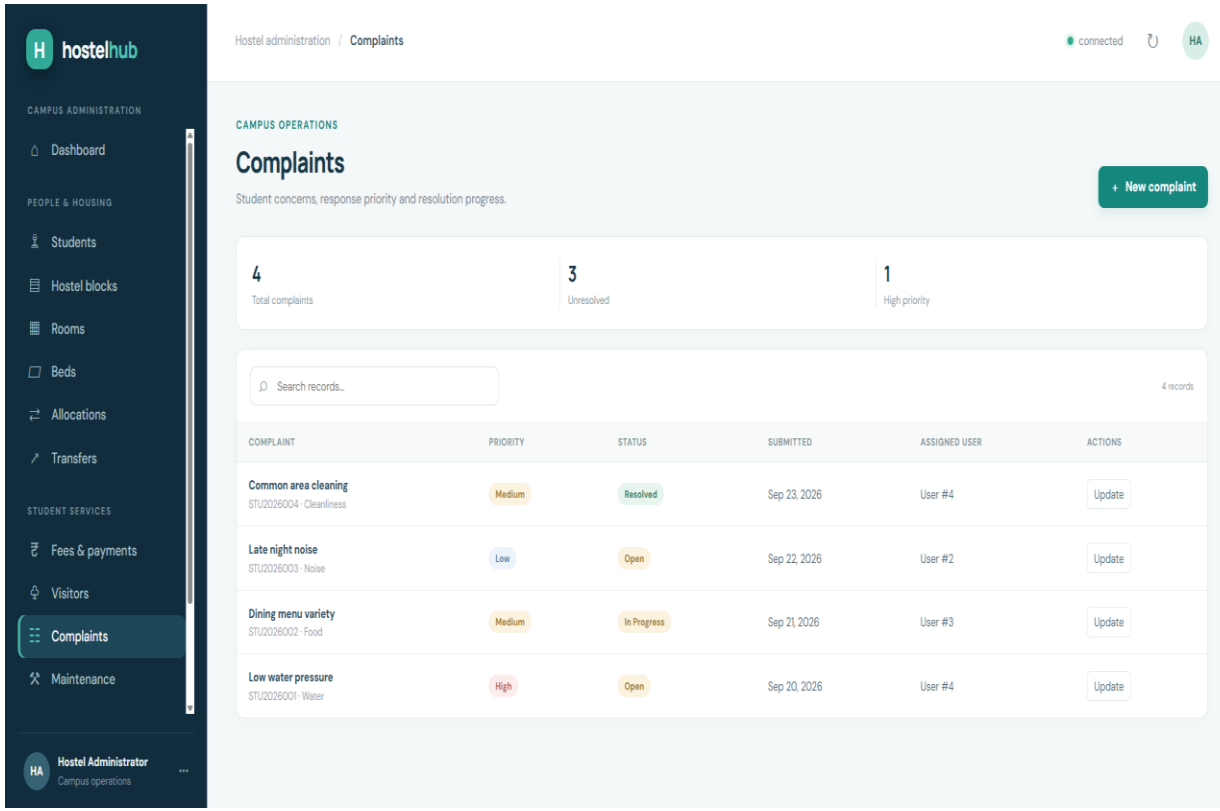


ROOMS

UI Screens: Fee Payment and Complaint

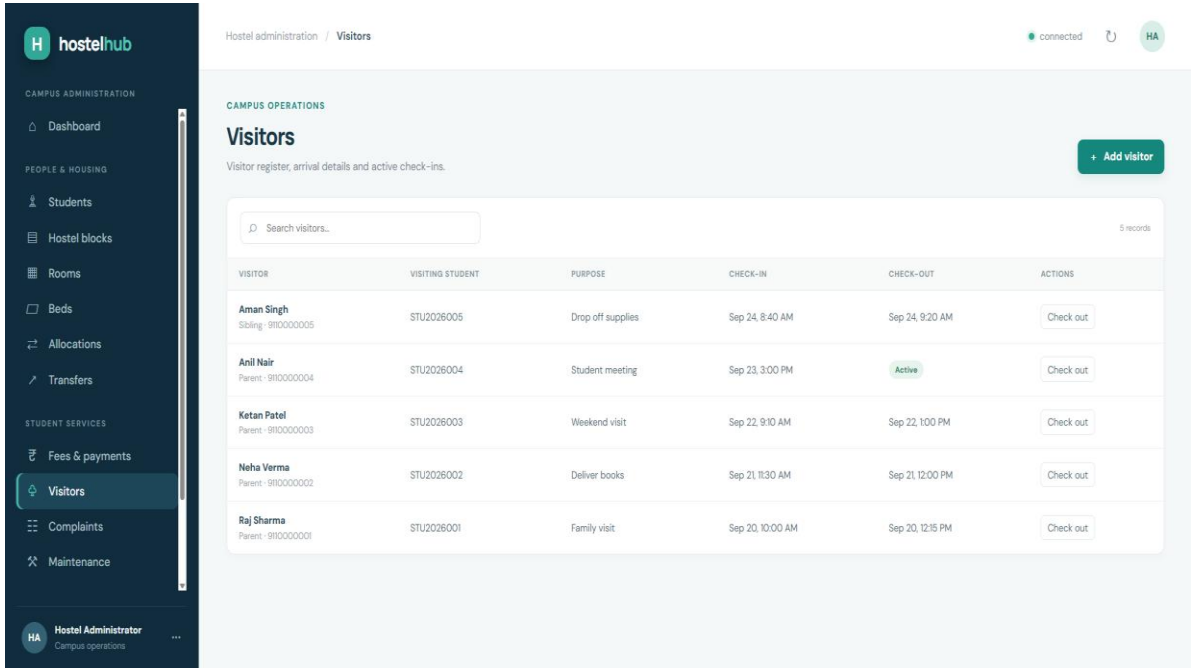


Fee Payment: view with status

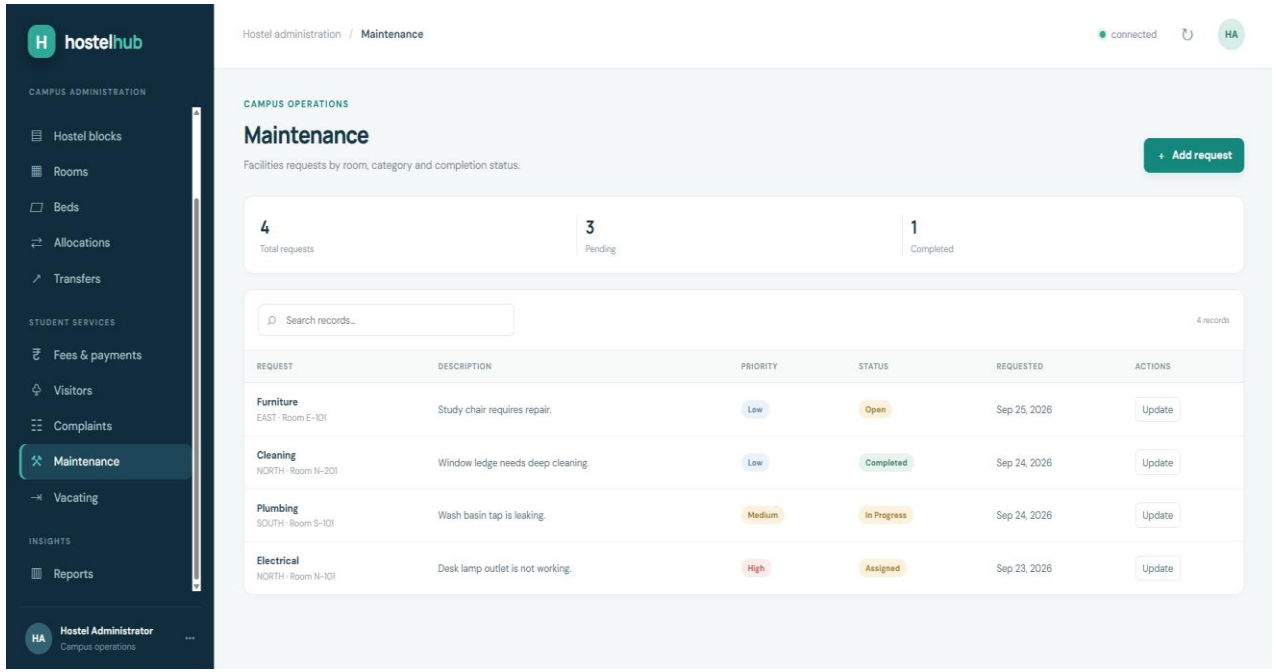


Complaint: view with status

UI Screens: Visitors&Maintanance

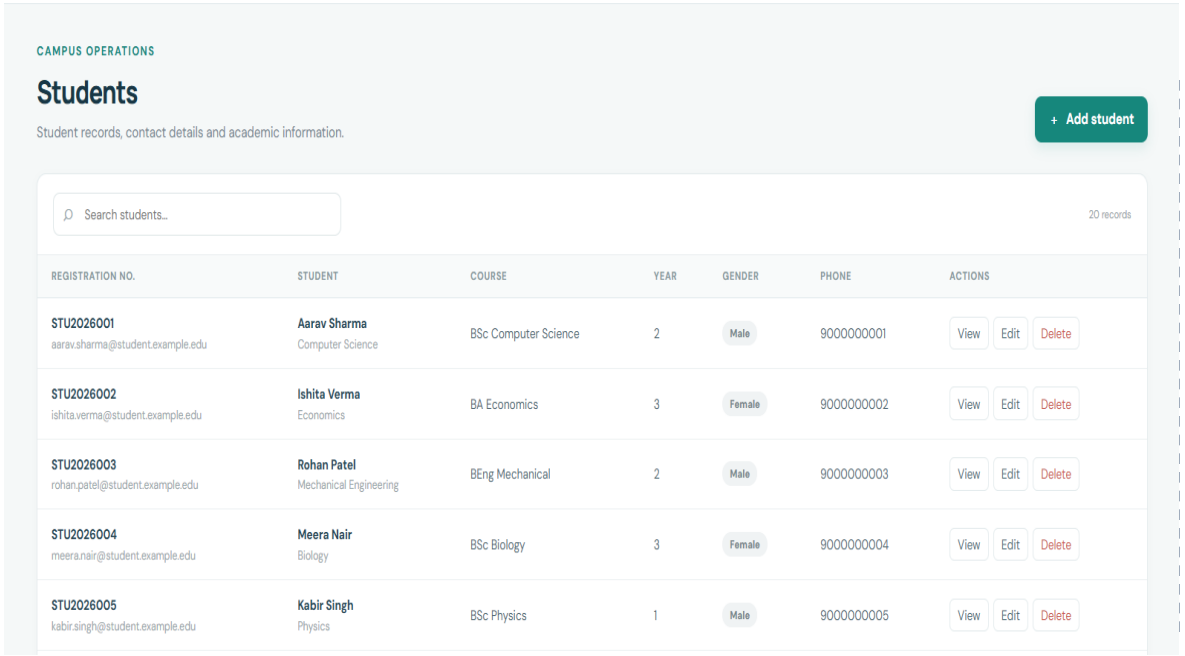


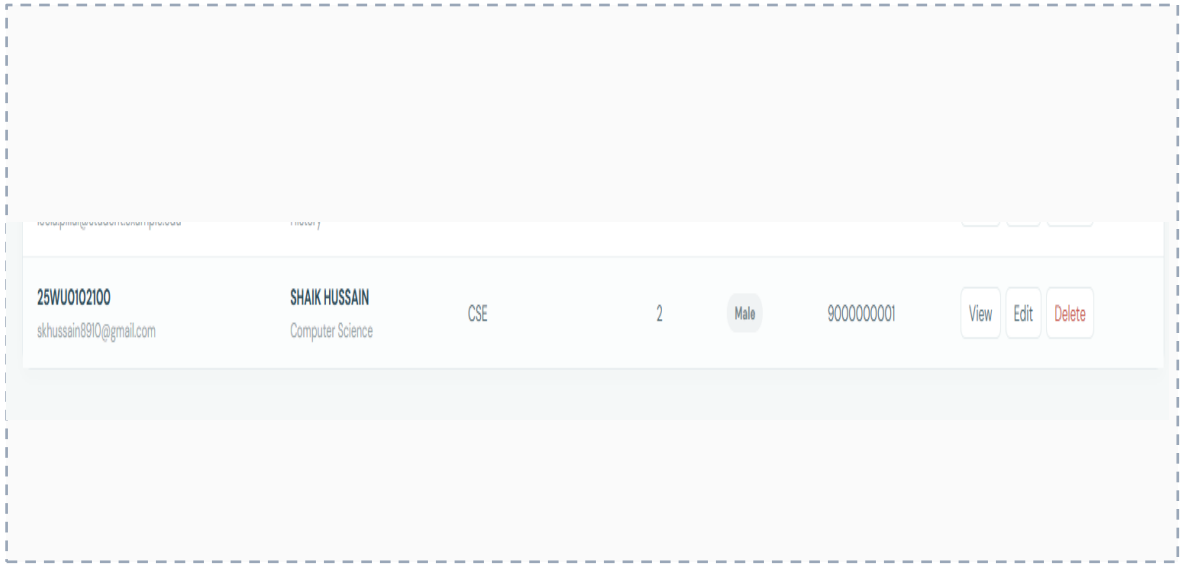
Visitor: view with entry/exit time



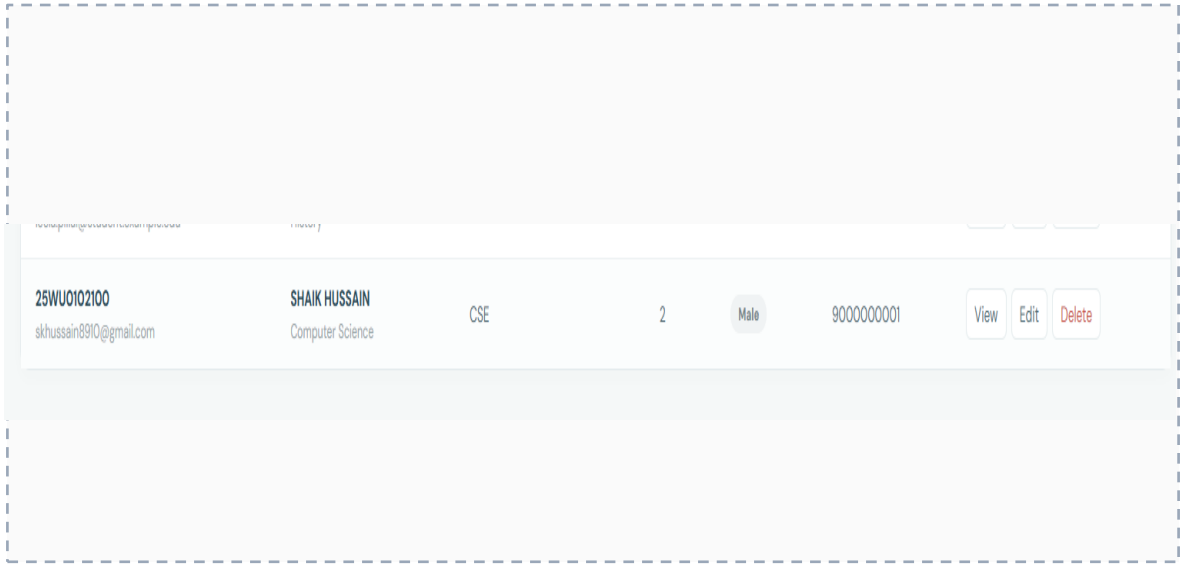
Maintanance

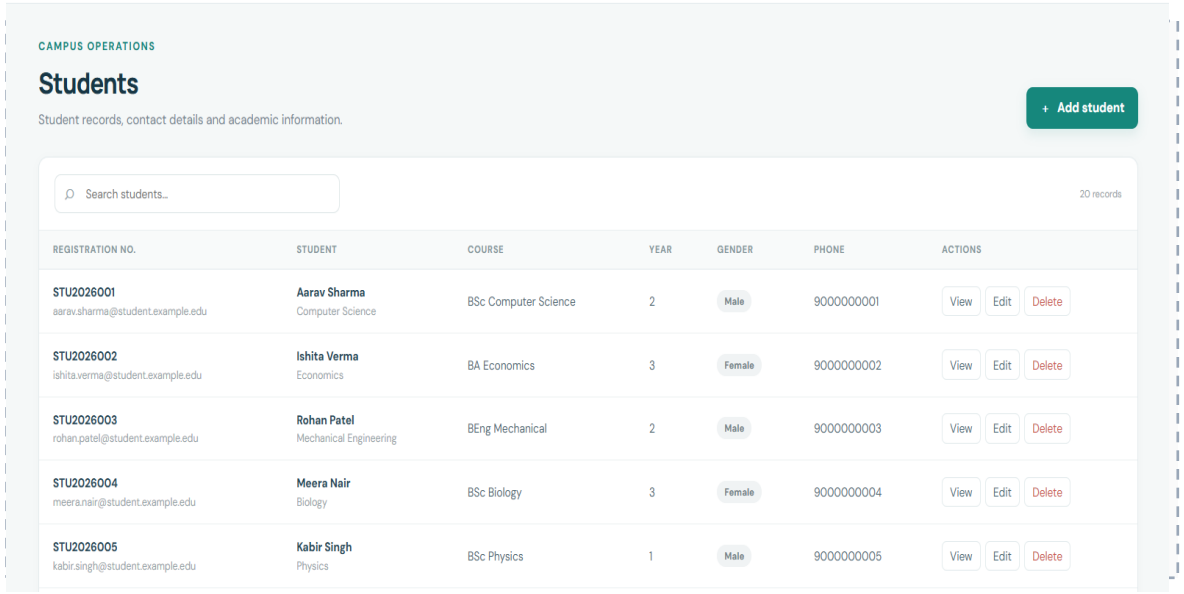
Insert Operation: Before and After





Delete Operation: Before and After





Each row has a Delete button; the row disappears from the table and from MySQL.

Conclusion and Future Work

- Normalised 8-table design with working insert, view and delete in every module
- Future: edit option, form validation, DD-MM-YYYY dates, login roles, fee reminders, dashboard charts
- Source code:https://github.com/HUSSAIN-134/HostelManagementSystem